Front-End Angular mini project

1. Title Page

- **Project Title**: *"Task Management and Productivity Tracker"*
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- **Course**: *Front-End Development with Angular*
- **Instructor Name**: *Thirupathy and Pallavi*
- **Institution**: *Christ University*
- **Date of Submission**: *10/03/2026*

2. Abstract

FocusFlow is a **Task Management and Productivity Tracking web application** developed using **Angular 17 and Angular Material**. The system enables users to efficiently manage daily tasks through a modern, responsive interface that supports task creation, modification, deletion, and progress tracking.

The application provides advanced productivity features such as **dashboard analytics for task insights, Kanban board workflow visualization, due-date countdown indicators, and bulk task operations**. Users can easily monitor their productivity using graphical charts and task statistics. The platform also supports **CSV export functionality**, allowing users to download task data for analysis or record keeping.

The project demonstrates modern Angular development practices including **modular architecture, reactive forms, custom pipes, Angular services, drag-and-drop interaction, and persistent theme management using local storage**. The interface is designed using Angular Material components to provide a consistent and responsive user experience.

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3. Objectives

a) To design and develop a **task management system** that helps users organize and track daily productivity activities.

b) To implement **full CRUD functionality** allowing users to create, view, update, and delete tasks.

c) To provide **visual productivity analytics** through dashboard charts and statistics.

d) To enable **workflow management using a Kanban board** with drag-and-drop functionality.

e) To highlight task urgency through **dynamic due-date countdown indicators**.

f) To improve productivity management through **bulk task actions for multiple task updates**.

g) To provide **data export capabilities** for downloading task records in CSV format.

h) To implement **dark and light theme switching** for better accessibility and user preference.

i) To maintain **data persistence using local storage** to ensure tasks remain available across sessions.

j) To demonstrate **best practices in Angular development**, including modular components, services, and reactive programming.

4. Scope of the Project

This project focuses on developing a **client-side task management web application** that allows users to efficiently organize and track tasks. The system includes task creation, editing, deletion, filtering, and visualization features.

The application provides advanced productivity tools such as **dashboard analytics, Kanban board workflow management, bulk task actions, due-date indicators, and CSV export functionality**. Additionally, the interface includes **dark mode support and persistent theme settings** to enhance usability.

The project does not include **backend database integration, multi-user collaboration features, or real-time cloud synchronization**. All task data is stored locally using the browser's **local storage API**. The application is designed as a standalone productivity tool demonstrating modern Angular development techniques.

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5. Tools & Technologies Used

The application is developed using **Angular 17 and TypeScript**, which provide a structured framework for building scalable web applications. Angular's component-based architecture enables modular development and reusable code structures.

The user interface is implemented using **Angular Material**, which provides a collection of responsive UI components based on Google's Material Design guidelines. Styling is implemented using **SCSS**, enabling modular and maintainable styling across components.

Task data management is handled through **Angular Services**, while persistent storage is implemented using the browser's **Local Storage API**. Drag-and-drop functionality for the Kanban board is implemented using **Angular CDK Drag and Drop**.

The application also utilizes **custom Angular pipes for dynamic data transformation**, along with routing mechanisms provided by the Angular Router for seamless navigation between application views.

Development and testing were performed using **Angular CLI, npm, Karma, and Jasmine**, which support efficient development workflows and testing practices.

6. Angular Project Structure Overview

```
focusflow-app/
│
├── src/
│   ├── app/
│   │   ├── components/
│   │   │   ├── dashboard/
│   │   │   ├── navbar/
│   │   │   ├── task-form/
│   │   │   ├── task-list/
│   │   │   └── kanban-board/
│   │   │
│   │   ├── pipes/
│   │   │   └── due-date.pipe.ts
│   │   │
│   │   ├── services/
│   │   │   └── task.service.ts
│   │   │
│   │   ├── shared/
│   │   │   └── material.module.ts
│   │   │
│   │   ├── app-routing.module.ts
│   │   └── app.module.ts
```

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```
│   ├── assets/
│   └── index.html
```

7. TypeScript Models and Interfaces

- Task Interface:

```
export interface Task {
id: string;
title: string;
description: string;
categoryId: string;
dueDate: string;
priority: 'low' | 'medium' | 'high';
status: 'pending' | 'in-progress' | 'completed';
progress: number;
createdAt: string;
updatedAt: string;
tags?: string[];
}
```

8. Angular Features Implemented :

Components & Data Binding:

The application is structured using multiple Angular components including the **Navbar, Task List, Task Form, Dashboard, and Kanban Board**. Each component contains its own HTML template, TypeScript logic, and SCSS styling to ensure modular design and code reusability. Angular data binding techniques are used extensively to dynamically render task data and synchronize UI elements with application state.

Services:

The application utilizes a centralized **TaskService** to manage task operations such as creating, updating, deleting, and retrieving tasks. This service also handles synchronization with the browser's **local storage**, ensuring that user data persists across sessions.

Routing:

Angular Router is used to enable navigation between different views such as the **dashboard page, task list view, and Kanban board view**. Routes are configured to ensure smooth navigation throughout the application.

Forms:

Reactive Forms are implemented to handle user input when creating or editing tasks. Form validation ensures that required fields such as task title, due date, and priority are correctly entered before submission.

9. CSS Styling Strategy

- Implemented modular styling using **component-specific SCSS files**.
- Utilized **Angular Material components** for consistent UI design.
- Implemented **responsive layouts using Flexbox and CSS Grid**.
- Added **dark and light theme support** using global theme classes.
- Used **priority color coding** to visually distinguish task importance.
- Applied **custom styling overrides for dark mode compatibility**.

10. Key Features Implemented

- Dashboard Analytics:
  The application includes a **dashboard analytics page** that provides users with a visual overview of task statistics. The dashboard displays statistical cards for total tasks, completed tasks, tasks in progress, and overdue tasks. Graphical visualizations including **completion percentage charts and status distribution charts** help users understand their productivity patterns.
- Kanban Board Workflow:
  A **Kanban board interface** is implemented to visualize task progress across workflow stages. Tasks are organized into **Pending, In Progress, and Completed** columns. Users can move tasks between these columns using drag-and-drop functionality, allowing intuitive task workflow management.
- Due-Date Countdown Indicators:
  Each task displays a **dynamic due-date badge** indicating the remaining time before the deadline. Tasks nearing their deadline show countdown labels, while overdue tasks are highlighted using red indicators. Completed tasks display a completion label to differentiate them from active tasks.
- Bulk Task Actions:
  The application supports **bulk task operations**, enabling users to select multiple tasks simultaneously using checkboxes. Users can perform actions such as marking multiple tasks as completed or deleting selected tasks. A floating toolbar appears when tasks are selected, providing quick access to these bulk actions.

- Dark Mode with Persistence:
  The application provides a **dark mode feature** that allows users to switch between light and dark themes. The selected theme preference is stored in the browser's **localStorage**, ensuring the theme remains active even after refreshing or reopening the application.

- CSV Export Functionality:
  The system includes an **Export CSV feature** that allows users to download all task data as a CSV file. The exported file contains details such as title, description, priority, status, due date, and tags. This functionality enables users to maintain external records of their tasks.

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11. Challenges Faced and Solutions

Challenge 1: Managing task state across multiple components.
Solution: Implemented a centralized TaskService to manage task data.

Challenge 2: Ensuring persistent data storage without a backend.
Solution: Used browser local storage to store task information.

Challenge 3: Implementing drag-and-drop workflow management.
Solution: Used Angular CDK Drag and Drop module.

Challenge 4: Maintaining consistent UI in dark mode.
Solution: Implemented global dark theme overrides and component-level styling fixes.

12. Learning Outcomes

Through this project, we gained practical experience in **Angular application architecture, component-based design, and TypeScript interface development**. We also improved our skills in implementing **responsive UI design using Angular Material**, managing application state using Angular services, and building dynamic user interfaces with reactive forms. The project provided valuable exposure to modern frontend development practices.

13. Future Enhancements

Future improvements may include **backend database integration**, user authentication systems, and real-time task synchronization across devices. Additional features such as **calendar views, team collaboration, notifications, and productivity analytics** could further enhance the application's functionality.

14. Conclusion

The FocusFlow Task Management application demonstrates the effectiveness of Angular in building modern, scalable web applications. By combining Angular's modular architecture with Material Design components, the project provides a responsive and user-friendly productivity management system. The application serves as a strong foundation for further enhancements and real-world deployment.

15. References

Angular Documentation – https://angular.io/docs
Angular Material Documentation – https://material.angular.io
TypeScript Documentation – https://www.typescriptlang.org/docs
Angular CLI Documentation – https://angular.io/cli
MDN Web Docs – https://developer.mozilla.org

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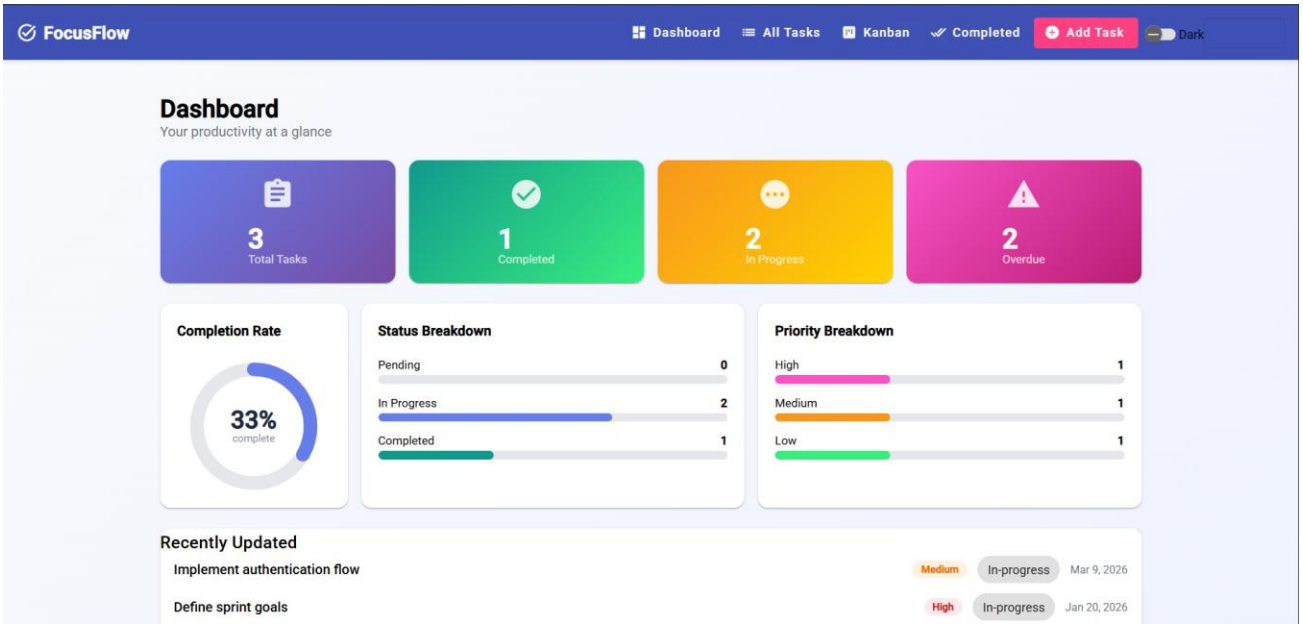
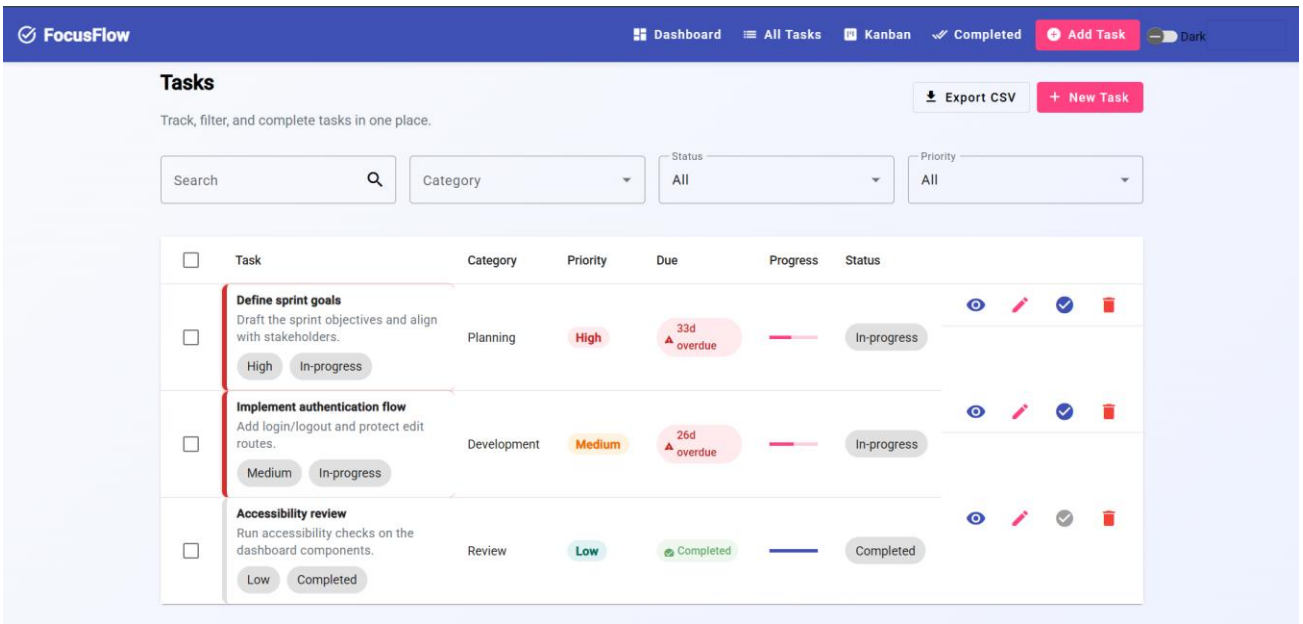
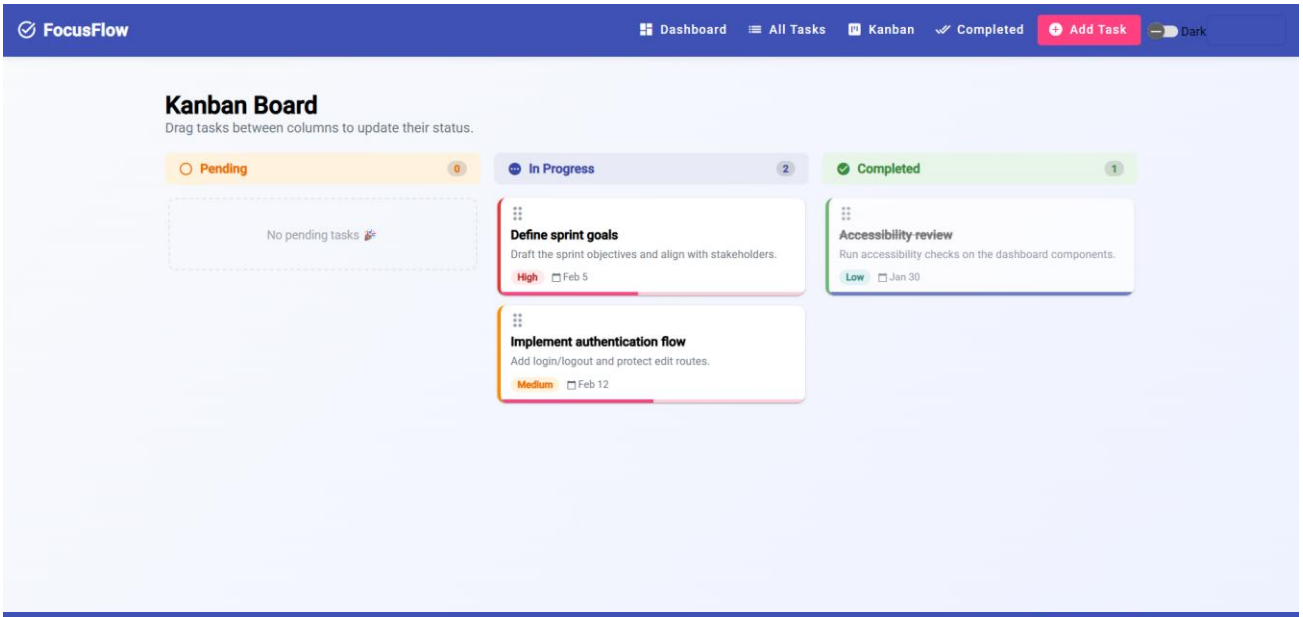
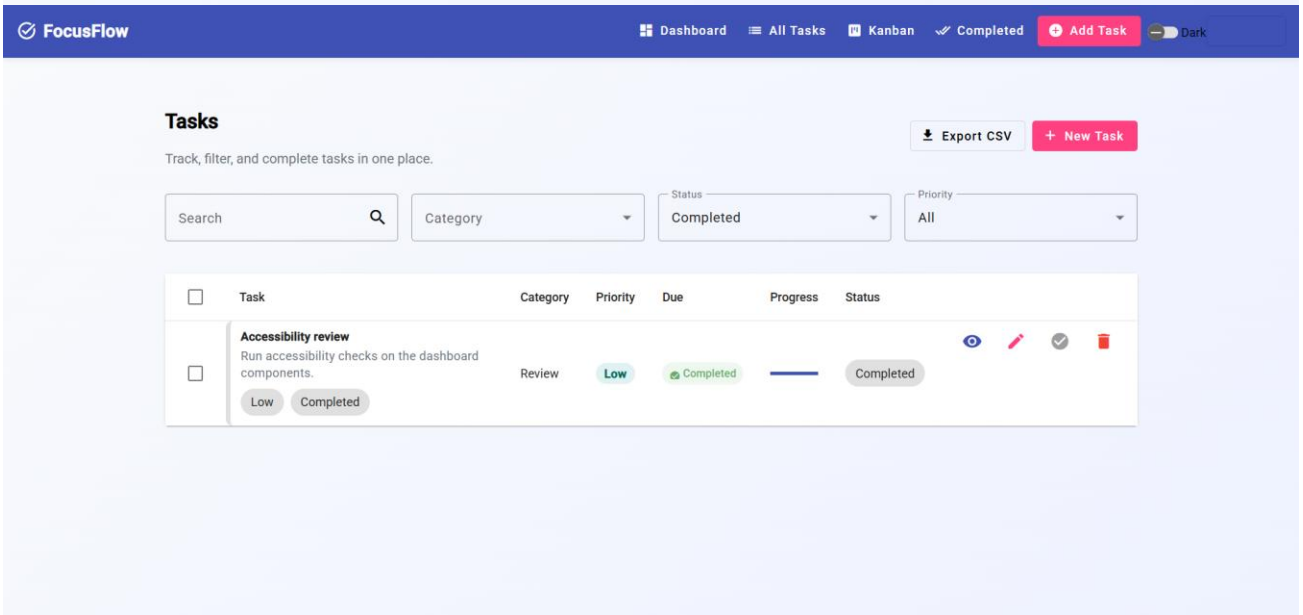
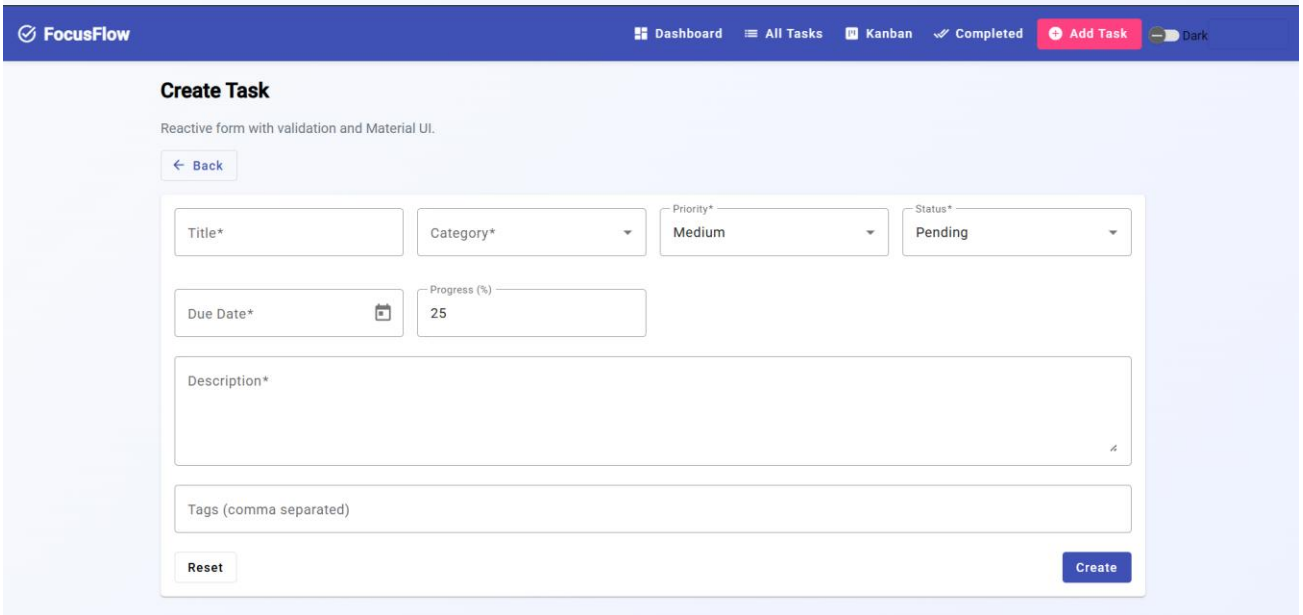
16. Screenshots

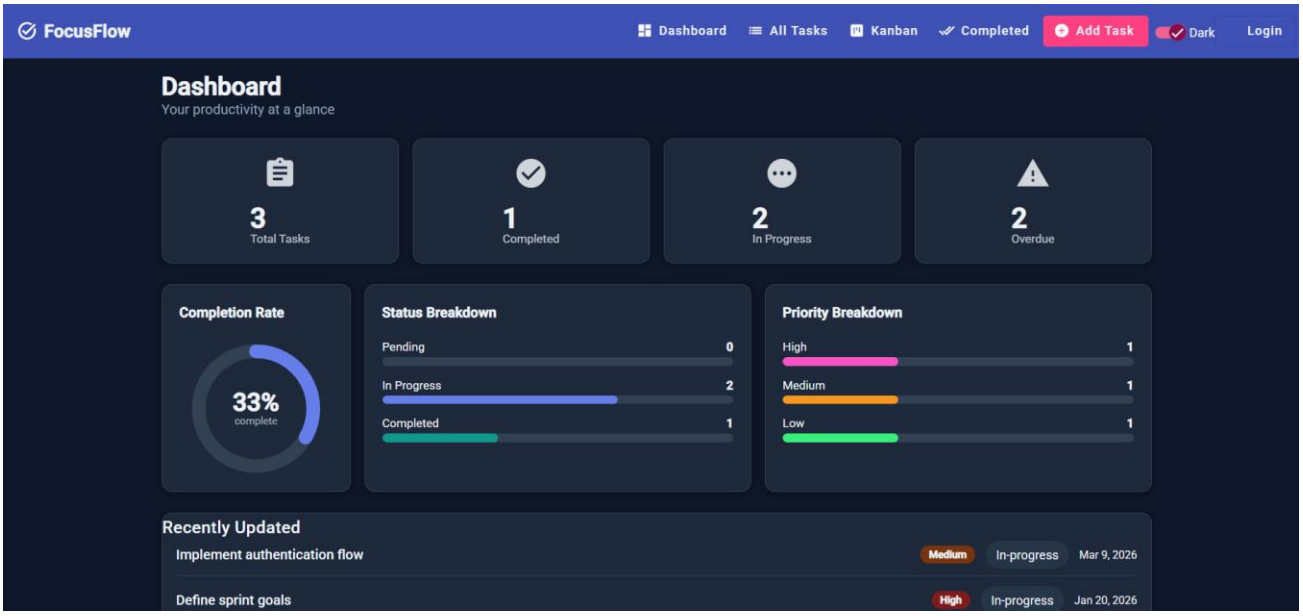


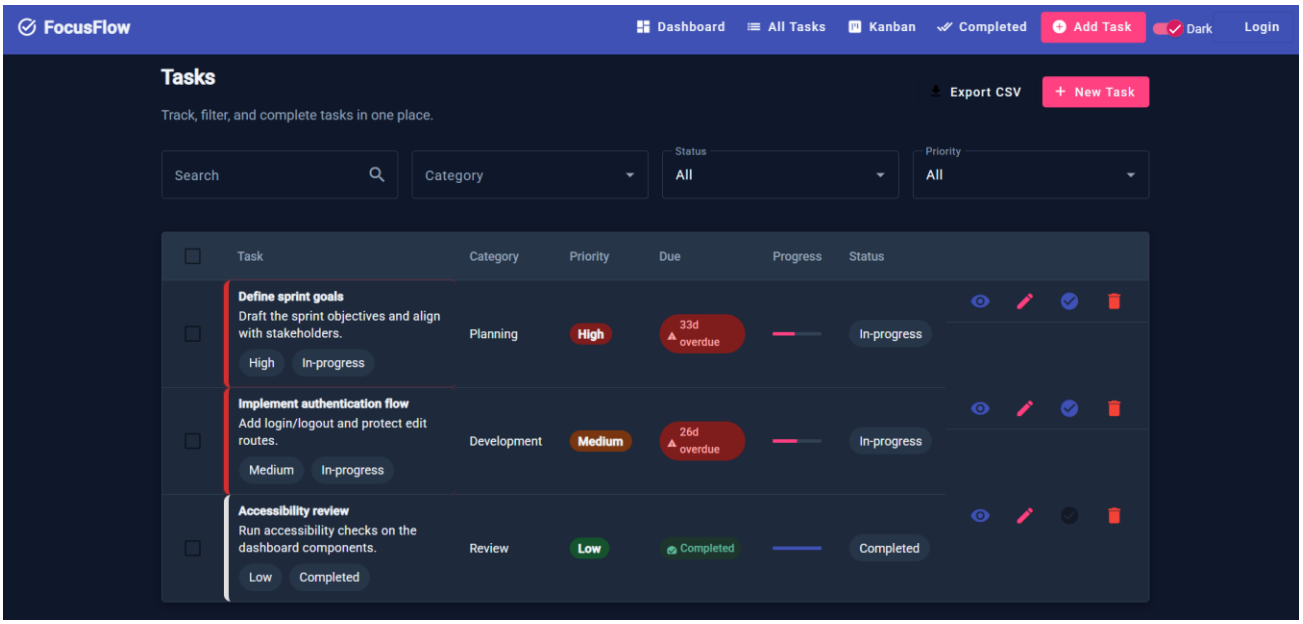


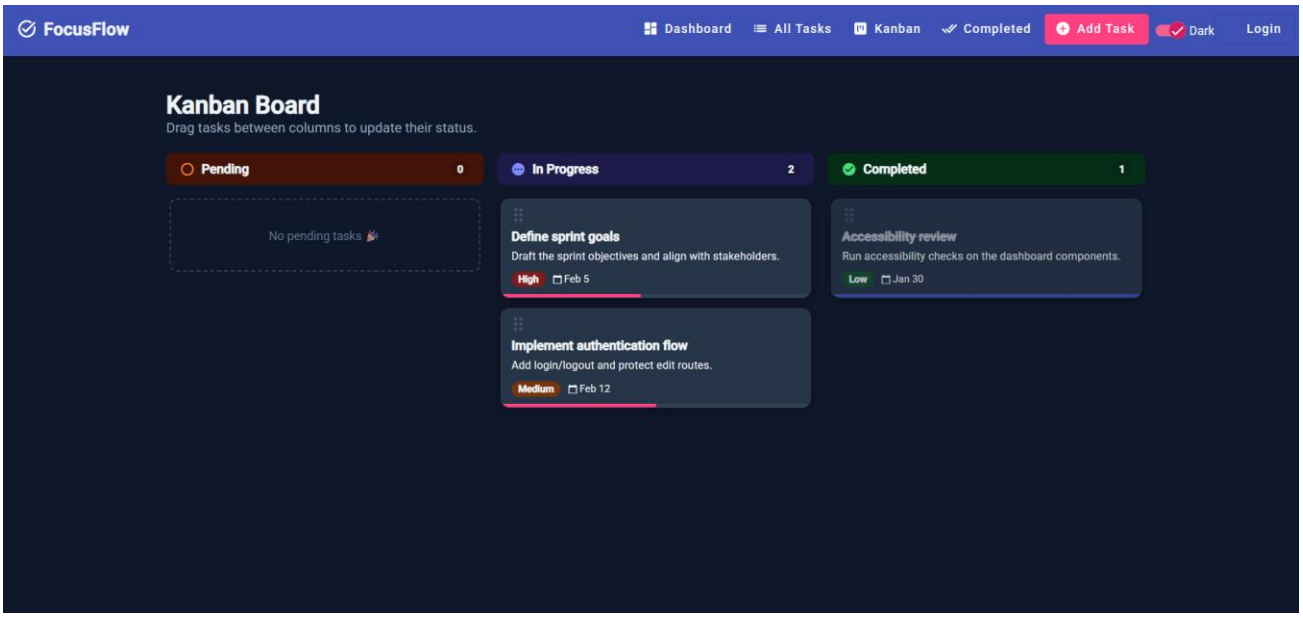

FocusFlow
Dashboard
All Tasks
Kanban
Completed
Add Task
Dark
Kanban Board
Drag tasks between columns to update their status.
Pending 0
In Progress 2
Completed 1
No pending tasks 🎉
Define sprint goals
Draft the sprint objectives and align with stakeholders.
High
Feb 5
Accessibility review
Run accessibility checks on the dashboard components.
Low
Jan 30
Implement authentication flow
Add login/logout and protect edit routes.
Medium
Feb 12

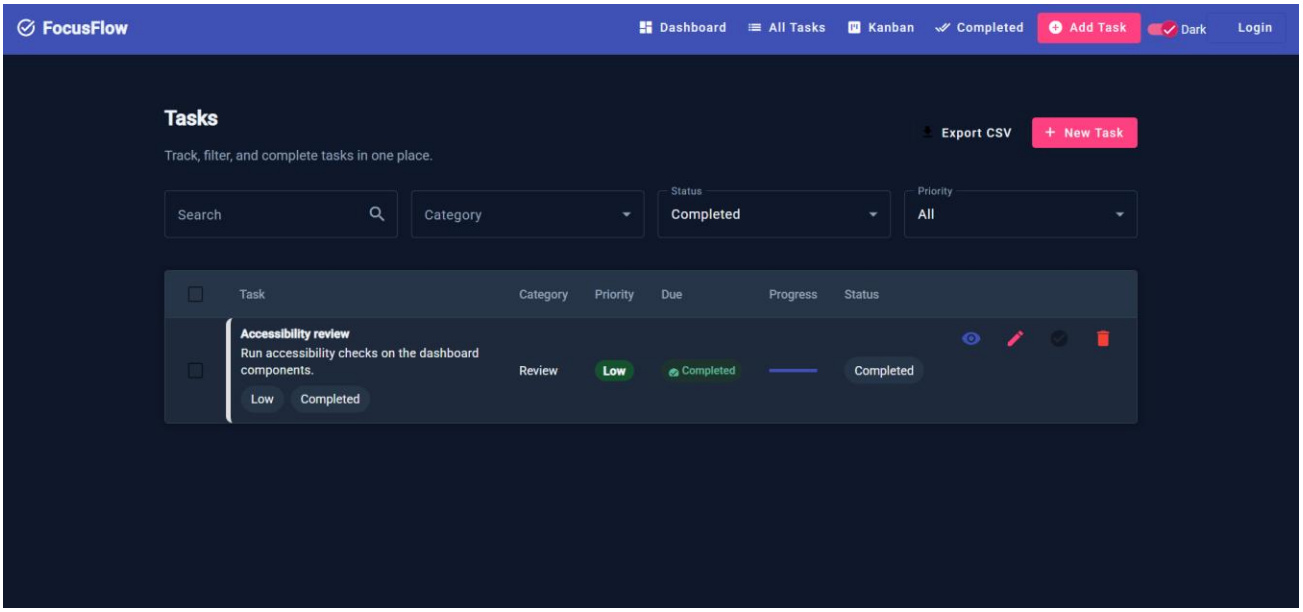

FocusFlow
Dashboard
All Tasks
Kanban
Completed
Add Task
Dark
Tasks
Track, filter, and complete tasks in one place.
Export CSV
New Task
Search
Category
Status
Completed
Priority
All
Task
Category
Priority
Due
Progress
Status
Accessibility review
Run accessibility checks on the dashboard components.
Low
Completed
Review
Low
Completed
Completed

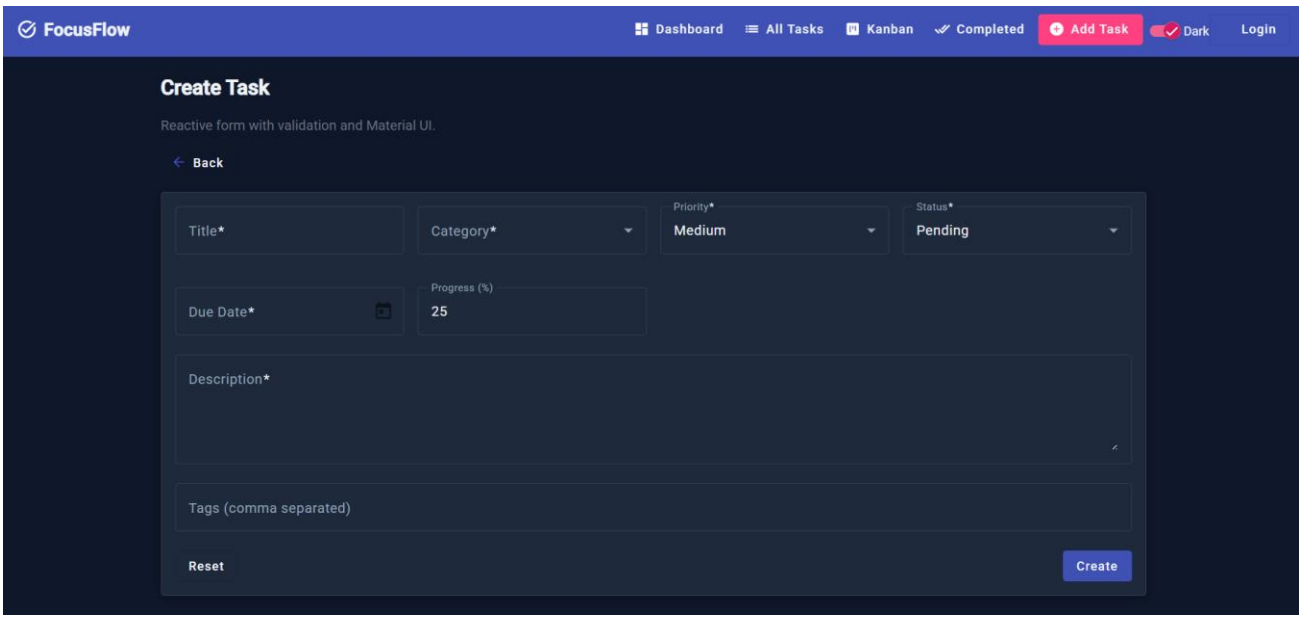

FocusFlow
Dashboard
All Tasks
Kanban
Completed
Add Task
Dark
Create Task
Reactive form with validation and Material UI.
Back
Title*
Category*
Priority*
Medium
Status*
Pending
Due Date*
Progress (%)
25
Description*
Tags (comma separated)
Reset
Create


FocusFlow
Dashboard
All Tasks
Kanban
Completed
Add Task
Dark
Login
Dashboard
Your productivity at a glance
3
Total Tasks
1
Completed
2
In Progress
2
Overdue
Completion Rate
33%
complete
Status Breakdown
Pending 0
In Progress 2
Completed 1
Priority Breakdown
High 1
Medium 1
Low 1
Recently Updated
Implement authentication flow
Medium
In-progress
Mar 9, 2026
Define sprint goals
High
In-progress
Jan 20, 2026


FocusFlow
Dashboard
All Tasks
Kanban
Completed
Add Task
Dark
Login
Tasks
Export CSV
New Task
Track, filter, and complete tasks in one place.
Search
Category
Status
All
Priority
All
Task
Category
Priority
Due
Progress
Status
Define sprint goals
Draft the sprint objectives and align with stakeholders.
High
In-progress
Planning
High
33d overdue
In-progress
Implement authentication flow
Add login/logout and protect edit routes.
Medium
In-progress
Development
Medium
26d overdue
In-progress
Accessibility review
Run accessibility checks on the dashboard components.
Low
Completed
Review
Low
Completed
Completed


FocusFlow
Dashboard
All Tasks
Kanban
Completed
Add Task
Dark
Login
Kanban Board
Drag tasks between columns to update their status.
Pending 0
No pending tasks
In Progress 2
Define sprint goals
Draft the sprint objectives and align with stakeholders.
High
Feb 5
Implement authentication flow
Add login/logout and protect edit routes.
Medium
Feb 12
Completed 1
Accessibility review
Run accessibility checks on the dashboard components.
Low
Jan 30





	A	B	C	D	E	F	G	H	I	J
1	Title	Description	Category	Priority	Status	Progress	Due Date	Tags	Created At	
2	Define sprint goals	Draft the sprint objectives and align with stakeholders.	cat-1	high	in-progress	45	2026-02-05T00:00:00.000Z	planning; sprint	2026-01-10T00:00:00.000Z	
3	Implement authentication flow	Add login/logout and protect edit routes.	cat-2	medium	in-progress	50	2026-02-12T00:00:00.000Z	security; frontend	2026-01-12T00:00:00.000Z	
4	Accessibility review	Run accessibility checks on the dashboard components.	cat-3	low	completed	100	2026-01-30T00:00:00.000Z	a11y; qa	2025-12-02T00:00:00.000Z	
5										
6										
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12										
13										

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